ProveIt research contribution

Integral Projections and Lattice Forms for Diophantine Consistency

Complete explicit quartics and exact certificates

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With AI assistance

Abstract

We construct and supply several ordinary integer-coefficient polynomials whose integer unsolvability is provably equivalent in ZFC to its syntactic consistency. The contribution is an exact compression of the pinned ProveIt direct proof-calculus constructions, using two complementary operations. First, an affine equation with a nonunit coefficient can eliminate a variable when a retained equation certifies that the rational restoration is integral. Second, a positive-definite integral Gram matrix with even diagonal can replace the diagonal sum-of-squares combiner. The principal result is a completely specified quartic in 240 unrestricted integer variables with 5,803 collected monomials and coefficient height 6,404. Its Gram form has only thirteen mixed residual products; a rank-eight block is an even unimodular lattice. A companion has 9,357 monomials and height 4,160. A simpler sum-of-squares companion has 9,217 monomials and height 4,512, improving both support and height over the inspected 240-variable predecessor's 21,482 and 6,144.

The paper prints every residual of the principal polynomial. The package also contains complete collected coefficient arrays, rational restoration certificates, exact positive-definiteness certificates, independent checkers, and immutable copies of the inherited source archives. The new transformations are proved here; the C222-to-ZFC calibration is explicitly inherited. We do not claim a global simplicity optimum, literature-wide priority, or a new end-to-end proof-assistant derivation. An additional, fully materialized eleven-axiom finite theory gives a precise research direction without being counted as a new compiled Diophantine endpoint.

Comparison snapshot: `7a8da414376a80b90ecf799f37ceb26fb9ac674e`.
All unknowns in the completed new equations range over $\mathbb{Z}$.

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1 What is obtained

The requested logical target is

$$\mathrm{ZFC} \vdash [\neg\exists \boldsymbol{x} \in \mathbb{Z}^n \; P(\boldsymbol{x}) = 0] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

Here consistency means the absence of a finite refutation in the repository's specified syntactic ZFC calculus. The equivalence does not assume consistency and does not prove either side separately.

The central completed polynomial in this report is

$$\boxed{Q_{240}(\boldsymbol{x}) = \sum_{i=0}^{176} r_i(\boldsymbol{x})^2 + \sum_{(i,j,\epsilon)\in E} \epsilon\, r_i(\boldsymbol{x}) r_j(\boldsymbol{x}) = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^{240}.} \tag{2}$$

Every r_i is a fixed quadratic integer polynomial, printed in Appendix A. The thirteen-element signed edge list E is printed in Section 4. Thus the displayed specification is a finite ordinary polynomial, with no variable exponent, factorial, binomial primitive, uninstantiated universal polynomial, or undecidable coefficient selection.

1.1 Exact measurements

The support count is the number of nonzero monomials after full collection; height is the maximum absolute collected coefficient. These are different from residual count, expression length, operation count, and coefficient bit complexity. All rows below have exact degree four and all their declared coordinates occur in the collected polynomial.

Equation	Variables	Residuals	Monomials	Height
Inspected old 240-variable form	240	177	21,482	6,144
S_{240}^{bal}: projected squares	240	177	5,477	15,107
S_{240}^{low}: projected squares	240	177	9,217	4,512
Q_{240}: principal Gram form	240	177	5,803	6,404
H_{240}: lower-height Gram form	240	177	9,357	4,160
F_{241}: factorial-free projection	241	177	5,141	7,968

The usual sum of 177 residual squares gives two useful projected forms. The Gram versions retain those residuals and add respectively thirteen and seven mixed products. They have the same zero sets as their respective sum-of-squares inputs.

The new sum-of-squares form S_{240}^{low} improves the inspected 240-variable Shared Factorial Error endpoint in both support and height:

$$21\,482 \longrightarrow 9\,217, \qquad 6\,144 \longrightarrow 4\,512.$$

The changes are reductions of approximately 57.09% and 26.56%. The lower-height Gram companion H_{240} reaches 4 160, while its support rises from 9 217 to 9 357. The principal Q_{240} uses fewer terms, 5 803, at height 6 404. These are distinct tradeoffs; no single one dominates every other profile.

1.2 Scope of the comparison

The pinned repository contains constructions with far fewer unknowns, including nine-positive-variable forms and a 37-integer quartic, using different coefficient architectures and circuit costs. The present work improves the direct numerical proof-calculus family with fully materialized small integer

coefficients. It is not a variable-count record across all representations. The meaningful novelty here is the certified use of integrality-aware projection together with a nondiagonal residual combiner on these complete equations.

The immediate comparison archives are `ConZFC_Shared_Factorial_Error_2026-10-25.zip` and `ConZFC_Factorial_Free_Binomial_CRT_2026-10-25.zip`, both in `docs/incoming` at the stated commit [1, 2, 3]. We inspected all seven incoming archives and the newest consolidated construction directories. Their earlier source histories are retained in the original archives rather than silently replaced.

2 The exact logical dependency

Let P_{src} denote one of the supplied predecessor polynomials. The source reports establish, by their ordinary arithmetic proofs and identified calibration interfaces,

$$\mathrm{ZFC} \vdash [\exists \boldsymbol{z} \in \mathbb{Z}^m \; P_{\mathrm{src}}(\boldsymbol{z}) = 0] \longleftrightarrow [\text{there is a finite ZFC refutation}]\,. \tag{3}$$

Their direct arithmetic route encodes derivations in the numerical calculus C222; it does not simulate K215 or instantiate Jones's universal equation. Its named logical interfaces are

`KEndpoint.c222_iff_zf_refutation`,

`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

The relevant source files are included for inspection and pinned to the same repository snapshot [4, 5]. The second interface is the constructible-universe equiconsistency transport from ZF to ZFC.

The new work proves explicit equivalences

$$\exists P_{\mathrm{new}} = 0 \;\longleftrightarrow\; \exists P_{\mathrm{src}} = 0$$

and, for a Gram companion, equality of the corresponding integer zero sets. Composing with (3) and negating gives (1). Every new argument uses elementary arithmetic of integers and rationals, finite polynomial identities, or an exact finite positive-definiteness certificate; all are formalizable in ZFC.

This is an ordinary mathematical proof composed with a named inherited theorem. The release does not contain a newly checked object-language ZFC derivation of the entire chain, nor a rebuilt transitive Lean/Rocq dependency closure. Finite coefficient checks authenticate the polynomials and transformations; they do not substitute for the unbounded source interpretation. The full predecessor reports and source proof closures are retained in the two original archives under `sources/`.

3 Projection through a nonunit coefficient

3.1 The general theorem

Ordinary gate elimination usually restricts a defining equation to $x = f(\boldsymbol{y})$ or $-x = f(\boldsymbol{y})$. A coefficient such as 2 is excluded because solving $2x = f(\boldsymbol{y})$ over $\mathbb{Q}$ can create spurious integer tuples $\boldsymbol{y}$ with a half-integral x. The missing condition can sometimes be supplied by a different equation that is already present.

Theorem 3.1 (Integral restoration)**.** *Let a finite integer-polynomial system in $(x, \boldsymbol{y})$ contain*

$$ax + L(\boldsymbol{y}) = 0, \qquad 0 \neq a \in \mathbb{Z},$$

and a different retained equation

$$b(\boldsymbol{y})x^d + h_{d-1}(\boldsymbol{y})x^{d-1} + \cdots + h_0(\boldsymbol{y}) = 0, \qquad d \geq 1,$$

with integer-polynomial coefficients. Assume $\gcd(a, b(\boldsymbol{y})) = 1$ *for each integer tuple under consideration. Substitute* $x = -L(\boldsymbol{y})/a$ *into every other equation, clear fixed rational denominators by nonzero constants, and delete the first equation. Then coordinate projection is a bijection between the old and new integer zero sets. The inverse inserts* $x = -L(\boldsymbol{y})/a$*. If* L *is affine and every old residual has degree at most two, every new residual still has degree at most two.*

Proof. The forward map follows by substitution. For the reverse map, start with an integer zero $\boldsymbol{y}$ of the new system and set $x = -L(\boldsymbol{y})/a \in \mathbb{Q}$. Every old equation vanishes, because its cleared version vanishes and each clearing constant is nonzero. Write $x = u/v$ in lowest terms, with $v > 0$. The affine equation implies $v \mid a$. Multiplying the retained degree-d equation by v^d shows $v \mid b(\boldsymbol{y})u^d$. Since $\gcd(u, v) = 1$, one has $v \mid b(\boldsymbol{y})$. The stated coprimality forces $v = 1$. Thus x is integral. The inverse is unique because $a \neq 0$. Finally, affine substitution does not increase the total degree of a polynomial. Removing the nonzero integer content of a residual also preserves its zero set. □

A monic retained equation, or one with leading coefficient -1, is enough. The proof is the standard rational-root argument. The contribution is the verified choice of pivots and integrality equations in the complete source systems. Without the second equation, deleting x from $2x - y = 0$ would incorrectly permit odd y.

3.2 The selected square witness

Both selected Shared Factorial Error parents have 241 unrestricted coordinates $z_0, \ldots, z_{240}$ and 178 residuals. Use zero-based row and variable indices. Source row 88 is

$$A(\boldsymbol{z}) - 2z_{39} = 0,$$

where the affine expression independent of z_{39} is

$$\begin{aligned} A(\boldsymbol{z}) = z_{103} - z_{10} - z_{11} - 4(z_{21} + z_{26} + z_{32} + z_{38} + z_{67}) \\ - 2(z_{24} + z_{29} + z_{31} + z_{34} + z_{36} + z_{42} + z_{46} + z_{68}). \end{aligned} \tag{4}$$

The different source row 23 is

$$z_{39} - (z_{11} + 2z_{38})^2 = 0. \tag{5}$$

Delete row 88 and substitute $z_{39} = A/2$ in every other row. After denominator clearing, (5) becomes

$$A - 2(z_{11} + 2z_{38})^2 = 0. \tag{6}$$

At any integer zero, A is therefore even and the rationally reconstructed $A/2$ is exactly the integer square $(z_{11} + 2z_{38})^2$. No new parity coordinate is needed.

All other residuals stay quadratic. Reindex retained coordinates in their old order:

$$x_j = z_j \quad (j < 39), \qquad x_j = z_{j+1} \quad (j \geq 39).$$

The certificate records each nonzero rational multiplier used to clear denominators and remove residual content. The projected balanced source has 5,477 collected monomials and height 15,107; the projected lower-height source has 9,217 and 4,512.

Eliminating z_{103} through the unit coefficient in the same row is a different projection. It expands many occurrences of a large sum. Eliminating the square coordinate z_{39} instead uses more arithmetic information but produces a much sparser 240-variable endpoint. This explains why a rule that considers only unit pivots can miss a useful dimension-preserving comparison.

3.3 A second application

The factorial-free principal source has 242 coordinates and 178 rows. Its row 88 has coefficient -2 at z_{67}, and retained row 52 says

$$z_{67} = 2 + 4z_{66} + 4z_{66}^2.$$

The same argument removes z_{67} and gives an explicit 241-variable quartic with 177 residuals, 5,141 monomials, and height 7,968. The predecessor's other 241-variable projection has 12,774 terms at the same height. The complete new system and every substitution are included in `projection/factorialfree241/`.

4 A positive lattice form in place of a sum of squares

4.1 The aggregation theorem

Theorem 4.1 (Even integral Gram aggregation)**.** *Let $r_0, \dots, r_{m-1} \in \mathbb{Z}[\boldsymbol{x}]$ and let H be a symmetric, positive-definite integer matrix with even diagonal. Then*

$$Q(\boldsymbol{x}) = \frac{1}{2}\,\boldsymbol{r}(\boldsymbol{x})^T H \boldsymbol{r}(\boldsymbol{x})$$

has integer coefficients and, for every real tuple $\boldsymbol{x}$,

$$Q(\boldsymbol{x}) = 0 \quad \Longleftrightarrow \quad r_i(\boldsymbol{x}) = 0 \text{ for every } i.$$

If each residual has degree at most two, then $\deg Q \le 4$.

Proof. The diagonal coefficient is $H_{ii}/2 \in \mathbb{Z}$; each mixed coefficient is $H_{ij} \in \mathbb{Z}$, because the two symmetric entries cancel the factor $1/2$. Positive definiteness gives $\boldsymbol{v}^T H \boldsymbol{v} > 0$ for every nonzero real vector $\boldsymbol{v}$. Apply this to $\boldsymbol{v} = \boldsymbol{r}(\boldsymbol{x})$. The degree bound follows by multiplying two residuals. □

The familiar sum of squares uses $H = 2I$. There is no need for that particular matrix if the only purpose of the final polynomial is to enforce simultaneous vanishing. Small off-diagonal coefficients can cancel large coefficients in the expanded polynomial.

This enlarges the aggregation family beyond integer changes of the residual row basis followed by an unweighted sum of squares. In formal residual coordinates, an integer sum of squares of integer linear forms has even mixed coefficients. Our selected form has odd mixed coefficients. This observation concerns the formal residual variables; it does not assert that the substituted polynomial admits no other sum-of-squares representation.

4.2 The complete thirteen-edge specification

Number the projected balanced residuals from 0 to 176. In addition to their ordinary squares, the principal polynomial contains

$$\begin{aligned} &- r_{12}r_{16} - r_{12}r_{19} - r_{12}r_{17} - r_{49}r_{51} - r_{19}r_{51} \\ &- r_6r_{21} - r_{17}r_{26} - r_{15}r_{21} + r_{14}r_{21} + r_{17}r_{19} \\ &- r_7r_{19} - r_7r_{17} + r_6r_{50}. \end{aligned} \tag{7}$$

Equations (2), (7), and Appendix A specify every coefficient of Q_{240}.

Only two blocks of H are nontrivial. On residual coordinates $(6, 14, 15, 21, 50)$ the first is

$$H_5 = \begin{pmatrix} 2 & 0 & 0 & -1 & 1 \\ 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 0 \\ -1 & 1 & -1 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

On coordinates $(7, 12, 16, 17, 19, 26, 49, 51)$ the second is

$$H_8 = \begin{pmatrix} 2 & 0 & 0 & -1 & -1 & 0 & 0 & 0 \\ 0 & 2 & -1 & -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 2 & 1 & -1 & 0 & 0 \\ -1 & -1 & 0 & 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 2 \end{pmatrix}.$$

The other 164 blocks are (2). Exact rational LDL^T elimination, in the displayed orders, gives

$$D_5 = \mathrm{diag}(2, 2, 2, \tfrac{1}{2}, 1), \qquad D_8 = \mathrm{diag}(2, 2, \tfrac{3}{2}, \tfrac{5}{6}, \tfrac{4}{5}, \tfrac{3}{4}, 2, \tfrac{1}{6}).$$

Every pivot is positive. The full unit lower triangular factors are supplied as rational numerator–denominator pairs in `gram/balanced/GRAM_CERTIFICATE.json`; the checker verifies $H = LDL^T$ entry by entry. This is an exact proof, independent of any floating-point eigenvalue estimate.

The two determinants are 4 and 1. The second block is an even, positive-definite, unimodular lattice of rank eight, so it is isometric to E_8 by the classical uniqueness theorem; Elkies gives a short author-hosted proof [6]. This identification is interesting but is not needed for the zero-set equivalence. The rational LDL^T certificate already proves everything required.

4.3 Measured effect and companion

The Gram change takes the balanced projected input from height 15,107 to 6,404, a reduction of 57.61%. Its support grows from 5,477 to 5,803, or 5.95%. The support/height tradeoff is explicit and does not change the dimension or degree.

The lower-height input has a separate seven-edge Gram form, supplied completely in `gram/lowheight/`. It changes 9,217 terms and height 4,512 to 9,357 and 4,160. Its two nontrivial blocks have orders seven and two, determinants 4 and 3, and exact positive pivots

$$(2, 2, \tfrac{3}{2}, 2, 1, \tfrac{4}{3}, \tfrac{1}{4}), \qquad (2, \tfrac{3}{2}).$$

That form is H_{240} in the results table.

Corollary 4.2 (Completed consistency equations)**.** *The principal polynomial Q_{240}, the lower-height companion H_{240}, both projected sum-of-squares polynomials, and the factorial-free 241-variable companion each satisfy* (1) *in their stated unrestricted integer domains.*

Proof. Theorem 3.1 transports integer solutions to the exact source equation. For either Gram companion, Theorem 4.1 first transports zeros to all its residuals. Apply the inherited source equivalence (3) and negate the resulting existential statement. □

5 Why the new equation is fully explicit

The main JSON record contains exactly 240 ordered variables and 177 quadratic residual lists. A sparse term

$$[c, [i, j, \ldots]]$$

denotes $cx_ix_j\cdots$; repeated indices denote repeated factors and an empty list denotes the constant c. All indices are zero-based. The additional list `gram_cross_terms` records the triples (i, j, ϵ) in (7).

`expanded_polynomial.json` contains every collected coefficient of the single ordinary quartic. The specification does not depend on a search run, an optimizer, a positivity oracle, or an unexpanded universal-polynomial parameter. Evaluators use integer arithmetic only. The fact that the equation encodes an undecidable consistency question does not prevent its coefficients, dimensions, degree, and evaluation at any given finite integer tuple from being calculated exactly.

The transformations also retain more than bare satisfiability. Projection and affine restoration are mutually inverse on the old and new integer zero sets. A Gram change preserves the entire real zero set of the residual conjunction. These stronger statements make the soundness argument particularly short and auditable.

6 Verification and its limits

The top-level `verify.py` checks the exact source-archive bindings, all new projection identities, both Gram certificates, every final collected coefficient, and the generated report data. It uses Python's standard library. The projection checker uses exponent-vector polynomials and does not import the producer, whose monomials are lists of indices. The Gram checker independently multiplies residuals, reconstructs each block, and checks the rational factorization.

The retained receipts include:

- full coefficient and integer-restoration checks for all three projections, including nine deliberately corrupted controls;
- independent checks of the thirteen- and seven-edge Gram forms, including rejection of a singular Gram matrix and a false factorization;
- exact regeneration of the projection JSON data and both Gram endpoints;
- a fresh replay of the inherited Shared Factorial Error release, including its parent arithmetic artifact checks and its printed equation check;
- separate reviews of the projection, Gram forms, and the inherited signed-radix/nearest-factorial proof boundary;
- exact evaluation-circuit and formula-tree checks for the eleven-axiom finite theory, including all 5,803 collected coefficients and five deliberately corrupted controls.

Required checks use explicit failures rather than Python assertions, and normal and optimized-interpreter runs are recorded. Independent reviews here mean separately performed reviews within an AI-assisted workflow, not outside human peer review.

6.1 A useful stopping certificate

For each projected system, form the matrix whose rows are the coefficients of the homogeneous quadratic parts of its 177 residuals. The supplied finite-field elimination gives rank 177 modulo the prime 1 000 003. A nonzero minor modulo a prime is a nonzero integer minor, so the rows are

independent over $\mathbb{Q}$. Therefore no nonzero constant rational linear combination of these residuals is affine.

This rules out exposing another affine row by ordinary constant row combinations in these fixed systems. It does not rule out a different projection order, polynomial multipliers, a changed source certificate, or a different proof calculus. It is a narrow and useful stopping result, not a lower bound on Diophantine representations of consistency.

6.2 What was not checked

We did not search for a zero, decide consistency, rebuild all historical Lean/Rocq dependencies, or prove a global optimum. No finite collection of sampled tuples would establish the universal logical equivalence. The checks establish literal algebra and finite certificates. The ordinary proofs establish the new universal transformations, and the credited source proofs provide the refutation interpretation.

7 A tested semantic redesign that did not improve the endpoint

The source theory can be altered safely by separating its full membership-vacuity family from a fixed-role seed family. One may also replace the four-parameter implicational axiom by three-parameter classical bases and express freshness using finitely many polynomial charts. However, reducing local schema arity does not by itself reduce the final Diophantine equation.

We compiled two explicit probes through the portable natural-number compiler. The three-parameter variant produced 246 natural coordinates and 201 residuals; the split four-parameter variant produced 226 and 178. Both exceed the inspected 219-natural-variable endpoint. The extra branches and constructor work outweigh the anticipated saving. The repository's incoming Digit Recovery experiments already contain a related three-parameter probe, so that direction is credited rather than claimed as unexplored.

These probes are not release endpoints and are not used in the proof of the completed quartics. Their measured failure is useful evidence for optimizing the entire compiled certificate, rather than optimizing only schema counts or the number of local parameters.

8 A different research route: a finite theory with one coded input

The preceding results are complete. The following construction is a proved logical transformation and a materialized finite-theory target for a new compiler; it does not yet supply a smaller materialized Diophantine equation.

Start with an explicitly given integer polynomial P and a finite arithmetic circuit that evaluates it. Consider the fixed finite language of commutative rings with identity, augmented by a binary pairing function π and unary projections ℓ, r. The eight function symbols are $0, 1, -, \ell, r, +, \cdot, \pi$, with respective arities $0, 0, 1, 1, 1, 2, 2, 2$. Let E contain the universal closures of the following eight ring identities:

$$\begin{aligned} (a+b)+c &= a+(b+c), & a+b &= b+a,\\ a+0 &= a, & a+(-a) &= 0,\\ (ab)c &= a(bc), & ab &= ba,\\ a1 &= a, & a(b+c) &= ab+ac, \end{aligned}$$

and the universal closures of

$$\ell(\pi(a,b)) = a, \qquad r(\pi(a,b)) = b.$$

Use a balanced binary pairing tree to encode a k-tuple as one element. The corresponding coordinate projections have depth at most $\lceil \log_2 k \rceil$ after padding to a power of two.

If the circuit for P has s stored wires, a second balanced tuple codes their values. Write $C_P(z, u)$ for the finite conjunction of its gate equations, where input values are projections of z and wire values are projections of u. Write $o(u)$ for the output projection. Define the single first-order sentence

$$\theta_P : \quad \forall z \, \exists u \, \big(C_P(z, u) \ \wedge \ o(u) \neq 0\big). \tag{8}$$

All conjunctions can be balanced; the logical form has only the two displayed outer quantifiers, though the finite matrix and its projection terms must still be counted.

Theorem 8.1 (Finite pairing-ring reduction)**.** *For every explicit integer polynomial P with its fixed finite arithmetic circuit, ZFC proves*

$$\mathrm{Con}(E + \theta_P) \quad \Longleftrightarrow \quad \neg \exists \boldsymbol{n} \in \mathbb{Z}^k \, P(\boldsymbol{n}) = 0.$$

Proof. If P has no integer zero, use the standard ring $\mathbb{Z}$. Choose a bijection $\pi : \mathbb{Z}^2 \to \mathbb{Z}$ with its two inverse coordinate functions. For any z, its projections are integers. The circuit has its uniquely computed integer wire values; pair those values into u. Its output is nonzero, so (8) holds. This is a model of the finite theory, hence the theory is consistent by first-order soundness.

Conversely, suppose an integer tuple $\boldsymbol{n}$ is a root. Form its ground paired term $z_{\boldsymbol{n}}$ using ordinary ring numerals, including additive inverses for negative integers. The projection identities reduce its input coordinates to those numerals. Instantiate (8) at $z_{\boldsymbol{n}}$. Its gate equations force, by a finite induction on the circuit, each projected wire value to equal its calculated integer numeral. Ring numeral normalization proves that the final value is 0. This contradicts the required nonzero output. These steps give a finite first-order refutation, uniformly constructible from the root and the finite circuit. The two arguments are formalizable in ZFC. □

Applying this theorem to a completed polynomial in this report gives another explicit finite theory equiconsistent with ZFC. It does not need a new arithmetized ZFC proof predicate, Robinson arithmetic, or a finite class-theory presentation.

8.1 A fully materialized instance

The directory `finite_theory/` supplies this theory for the principal Q_{240}, including its complete arithmetic directed acyclic graph and all eleven literal first-order syntax trees. Integer constants are constructed from 0 and 1 through binary addition chains and negation. Every occurrence of a tuple projection is expanded into compositions of ℓ and r; every conjunction is a literal balanced binary tree. There are no unexpanded tuple macros or gate-specific function symbols in the artifact.

Measurement	Exact value
Arithmetic gates	2,736
Addition / multiplication / negation gates	1,200 / 1,040 / 254
Input gates / zero gates / one gates	240 / 1 / 1
Arithmetic graph depth (edges)	23
Input / gate tuple projection depths (edges)	8 / 12
Syntax nodes in $\theta_{Q_{240}}$	107,256
Syntax nodes in the eleven-sentence theory	107,353
Maximum functional syntax depth (edges)	28
Formula-only depth / maximum term depth (edges)	15 / 13
Complete theory JSON size (bytes)	697,036

The sentence $\theta_{Q_{240}}$ has exactly two quantifier nodes and quantifier depth two. Counting the ten background sentences as well, the theory has twenty-two quantifier nodes and maximum quantifier depth three. The syntax-node counts include repeated projection terms; whole-theory counts are sums and depths are maxima over its eleven sentences. These are measurements of the functional language, not measurements of a relational translation or a numerical formula code.

An independent checker expands the complete arithmetic graph and, separately, the residual Gram form, comparing all 5,803 coefficients. It checks every syntax node, symbol arity, bound variable, background axiom, gate equation, conjunction, and metric. The builder regenerates the five mathematical JSON files byte for byte. This supplies a concrete compilation input rather than merely a schematic sentence.

8.2 The remaining compilation problem

The engineering opportunity is to compile refutations of this finite theory with a fixed logical kernel, moving the particular polynomial into fixed finite axiom-code data. Balanced tuple coding avoids the naive prefix of one quantifier per input or intermediate wire. It is not free compression: a complete implementation must relationalize the function language, supply the exact axiom codes, account for their bit lengths, and prove the new source-to-calculus calibration. The present Cantor-style formula coding can make a deep encoded sentence astronomically large. No improved variable count, degree, or practical coefficient size is asserted for this route until those costs are measured and the final polynomial is emitted.

9 Files and incorporation

The package is intended as a new contribution directory; it does not overwrite an existing numbered construction. The principal files are:

Path	Purpose
`gram/balanced/`	Principal Q_{240}, full coefficients and LDL^T certificate
`gram/lowheight/`	Lower-height H_{240} with its complete certificate
`projection/`	Three exact projections, source copies, restoration maps and rank certificates
`finite_theory/`	Complete eleven-axiom syntax trees, 2,736-gate circuit and independent checker
`sources/`	Two immutable parent archives and pinned calibration excerpts
`review/`	Actual verification receipts and scoped independent reviews
`verify.py`	Offline verification entry point
`evaluate.py`	Exact evaluation at a supplied integer tuple
`build_report.py`	Rebuild this standalone LaTeX source and PDF
`SOURCE_LEDGER.json`	Source archive hashes, commit, paths and provenance
`MANIFEST.sha256`	Integrity hashes of the delivered payload

Use Python 3.10 or later:

```
python3 -B verify.py
python3 -B -O verify.py
python3 -B verify.py --rebuild
python3 -B evaluate.py --profile gram-balanced point.json
python3 -B build_report.py
```

The last command additionally requires a conventional pdfLaTeX installation with the packages listed in the source preamble. PDF metadata and TeX-engine versions can change PDF bytes while the mathematical JSON data remain identical.

A Every residual of the principal equation

All $x_0, \dots, x_{239}$ range independently over $\mathbb{Z}$. The following finite list defines $r_0, \dots, r_{176}$. Together with the thirteen mixed terms in (7), this is the complete polynomial Q_{240}. The formulas are generated from the canonical JSON and checked against it.

$$
\begin{aligned}
r_{0} &= -x_{3}^{2} + x_{4} \\
r_{1} &= x_{0} - x_{0}x_{2} - x_{2} + x_{9} \\
r_{2} &= -1 + x_{1} - x_{2}x_{7} - x_{3} \\
r_{3} &= -1 + x_{1} - x_{8}x_{9} \\
r_{4} &= 1 + x_{4} - x_{5}x_{20} + x_{18} \\
r_{5} &= -1 - x_{10}^{2} - 2x_{10}x_{11} - x_{11}^{2} + x_{21} \\
r_{6} &= -5 - 10x_{10} - 4x_{10}^{2} - 16x_{10}x_{11} - 8x_{10}x_{21} - 20x_{11} - 16x_{11}^{2} \\
&\quad - 16x_{11}x_{21} - 10x_{21} - 4x_{21}^{2} + x_{22} \\
r_{7} &= -2 + 2x_{10} - x_{10}^{2} - 4x_{10}x_{22} - 4x_{22} - 4x_{22}^{2} + x_{23} \\
r_{8} &= 1 + 6x_{10} - 36x_{10}^{2} - 24x_{10}x_{23} + 2x_{23} - 4x_{23}^{2} + x_{24} \\
r_{9} &= -2 - x_{11} - x_{11}^{2} - 2x_{11}x_{12} - x_{12} - x_{12}^{2} + x_{25} \\
r_{10} &= -9 + 7x_{10} - x_{10}^{2} + 4x_{10}x_{12} - 4x_{10}x_{25} - 12x_{12} - 4x_{12}^{2} \\
&\quad + 8x_{12}x_{25} + 12x_{25} - 4x_{25}^{2} + x_{26} \\
r_{11} &= -1 + x_{10} - x_{10}^{2} - 2x_{10}x_{12} + x_{12} - x_{12}^{2} + x_{27} \\
r_{12} &= -2x_{10} - 36x_{10}^{2} - 48x_{10}x_{11} - 72x_{10}x_{12} - 24x_{10}x_{21} \\
&\quad - 24x_{10}x_{27} - 4x_{11} - 16x_{11}^{2} - 48x_{11}x_{12} - 16x_{11}x_{21} \\
&\quad - 16x_{11}x_{27} - 36x_{12}^{2} - 24x_{12}x_{21} - 24x_{12}x_{27} - 2x_{21} \\
&\quad - 4x_{21}^{2} - 8x_{21}x_{27} - 4x_{27}^{2} + x_{28} \\
r_{13} &= -4x_{26}^{2} - 8x_{26}x_{28} - 4x_{28}^{2} + x_{29} \\
r_{14} &= -6 - 8x_{10} - 4x_{10}^{2} - 8x_{10}x_{11} + 8x_{10}x_{25} + 8x_{10}x_{27} - 8x_{11} \\
&\quad - 4x_{11}^{2} + 8x_{11}x_{25} + 8x_{11}x_{27} + 8x_{25} - 4x_{25}^{2} - 8x_{25}x_{27} \\
&\quad + 8x_{27} - 4x_{27}^{2} + x_{30} \\
r_{15} &= 1 - 6x_{10} - 4x_{10}^{2} + 16x_{10}x_{11} + 8x_{10}x_{21} + 16x_{10}x_{27} \\
&\quad + 8x_{10}x_{30} + 4x_{11} - 16x_{11}^{2} - 16x_{11}x_{21} - 32x_{11}x_{27} \\
&\quad - 16x_{11}x_{30} + 2x_{21} - 4x_{21}^{2} - 16x_{21}x_{27} - 8x_{21}x_{30} + 8x_{27} \\
&\quad - 16x_{27}^{2} - 16x_{27}x_{30} + 4x_{30} - 4x_{30}^{2} + x_{31} \\
r_{16} &= -10 - 5x_{10} - x_{10}^{2} - 8x_{10}x_{11} - 12x_{10}x_{12} - 4x_{10}x_{25} - 24x_{11} \\
&\quad - 16x_{11}^{2} - 48x_{11}x_{12} - 16x_{11}x_{25} - 36x_{12} - 36x_{12}^{2} \\
&\quad - 24x_{12}x_{25} - 12x_{25} - 4x_{25}^{2} + x_{32} \\
r_{17} &= -2 - 16x_{10} - 16x_{10}^{2} - 8x_{10}x_{11} - 48x_{10}x_{12} - 16x_{10}x_{27} \\
&\quad - 3x_{11} - x_{11}^{2} - 12x_{11}x_{12} - 4x_{11}x_{27} - 24x_{12} - 36x_{12}^{2} \\
&\quad - 24x_{12}x_{27} - 8x_{27} - 4x_{27}^{2} + x_{33}
\end{aligned}
$$

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\begin{aligned}
r_{18} &= 2 - 4x_{32}^2 - 8x_{32}x_{33} - 4x_{33}^2 + x_{34} \\
r_{19} &= -12 - 19x_{10} - 4x_{10}^2 - 24x_{10}x_{11} + 8x_{10}x_{21} + 8x_{10}x_{25} \\
&\quad - 46x_{11} - 36x_{11}^2 + 24x_{11}x_{21} + 24x_{11}x_{25} - 2x_{12} + 16x_{21} \\
&\quad - 4x_{21}^2 - 8x_{21}x_{25} + 14x_{25} - 4x_{25}^2 - x_{27} + x_{35} \\
r_{20} &= -3 - 6x_{10} - 4x_{12} - 2x_{27} - 4x_{35} - 4x_{35}^2 + x_{36} \\
r_{21} &= 15 + 36x_{10} + 3x_{10}^2 - 16x_{10}x_{11} - 8x_{10}x_{21} + 28x_{10}x_{22} \\
&\quad + 36x_{11} - 48x_{11}^2 - 48x_{11}x_{21} + 64x_{11}x_{22} + 18x_{21} - 12x_{21}^2 \\
&\quad + 32x_{21}x_{22} + 15x_{22} - 4x_{22}^2 + x_{37} \\
r_{22} &= 1 + 6x_{10} - 36x_{10}^2 - 24x_{10}x_{37} + 2x_{37} - 4x_{37}^2 + x_{38} \\
r_{23} &= -x_{10} - x_{11} - 2x_{11}^2 - 8x_{11}x_{38} - 4x_{21} - 2x_{24} - 4x_{26} - 2x_{29} \\
&\quad - 2x_{31} - 4x_{32} - 2x_{34} - 2x_{36} - 4x_{38} - 8x_{38}^2 - 2x_{41} - 2x_{45} \\
&\quad - 4x_{66} - 2x_{67} + x_{102} \\
r_{24} &= -6 - 2x_{10} - 4x_{11} - 14x_{21} - 16x_{21}^2 + x_{39} \\
r_{25} &= -4 + 6x_{11} - 36x_{11}^2 - 48x_{11}x_{12} - 24x_{11}x_{25} - 24x_{11}x_{39} \\
&\quad + 4x_{12} - 16x_{12}^2 - 16x_{12}x_{25} - 16x_{12}x_{39} + 2x_{25} - 4x_{25}^2 \\
&\quad - 8x_{25}x_{39} + 4x_{39} - 4x_{39}^2 + x_{40} \\
r_{26} &= 6x_{10} - 36x_{10}^2 - 48x_{10}x_{12} - 24x_{10}x_{27} - 24x_{10}x_{40} + 4x_{12} \\
&\quad - 16x_{12}^2 - 16x_{12}x_{27} - 16x_{12}x_{40} + 2x_{27} - 4x_{27}^2 - 8x_{27}x_{40} \\
&\quad + 4x_{40} - 4x_{40}^2 + x_{41} \\
r_{27} &= -x_{12}^2 - 2x_{12}x_{13} - x_{13}^2 + x_{42} \\
r_{28} &= 2x_{10} - x_{10}^2 - 4x_{10}x_{12} - 4x_{10}x_{42} + 4x_{12} - 4x_{12}^2 \\
&\quad - 8x_{12}x_{42} + 4x_{42} - 4x_{42}^2 + x_{43} \\
r_{29} &= -35 - 48x_{12} - 16x_{12}^2 + 16x_{12}x_{25} + 16x_{12}x_{27} + 24x_{25} \\
&\quad - 4x_{25}^2 - 8x_{25}x_{27} + 24x_{27} - 4x_{27}^2 + x_{44} \\
r_{30} &= -4x_{12} - 16x_{27}^2 - 32x_{27}x_{42} - 16x_{27}x_{43} - 16x_{27}x_{44} - 4x_{42} \\
&\quad - 16x_{42}^2 - 16x_{42}x_{43} - 16x_{42}x_{44} - 2x_{43} - 4x_{43}^2 - 8x_{43}x_{44} \\
&\quad - 4x_{44}^2 + x_{45} \\
r_{31} &= -16x_{10}^2 + 8x_{10}x_{12} + 32x_{10}x_{27} + 16x_{10}x_{30} + x_{12} - x_{12}^2 \\
&\quad - 8x_{12}x_{27} - 4x_{12}x_{30} - 16x_{27}^2 - 16x_{27}x_{30} - 4x_{30}^2 + x_{46} \\
r_{32} &= -1 - 8x_{12} - 4x_{12}^2 + 8x_{12}x_{25} + 8x_{25} - 4x_{25}^2 + x_{47} \\
r_{33} &= -1 - 9x_{12}^2 + 24x_{12}x_{25} + 12x_{12}x_{47} - 16x_{25}^2 - 16x_{25}x_{47} \\
&\quad - 4x_{47}^2 + x_{48} \\
r_{34} &= 1 + x_{11} - x_{11}^2 - 4x_{11}x_{12} - 4x_{11}x_{48} - 4x_{12}^2 - 8x_{12}x_{48} \\
&\quad - 4x_{48}^2 + x_{49} \\
r_{35} &= -1 - 4x_{49} - 4x_{49}^2 + x_{50} \\
r_{36} &= -1 - x_{12}^2 - 8x_{12}x_{49} - 4x_{12}x_{50} - 16x_{49}^2 - 16x_{49}x_{50} \\
&\quad - 4x_{50}^2 + x_{51} \\
r_{37} &= 2 + x_{11} - x_{11}^2 - 4x_{11}x_{12} - 4x_{11}x_{51} - 4x_{12}^2 - 8x_{12}x_{51} \\
&\quad - 4x_{51}^2 + x_{52} \\
r_{38} &= -4x_{46}^2 - 8x_{46}x_{52} - 4x_{52}^2 + x_{53}
\end{aligned}
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\begin{aligned}
r_{39} = & -9+7x_{10}-x_{10}^2-8x_{10}x_{46}-4x_{10}x_{53}+24x_{46}-16x_{46}^2 \\
& -16x_{46}x_{53}+12x_{53}-4x_{53}^2+x_{54} \\
r_{40} = & -1-4x_{54}-4x_{54}^2+x_{55} \\
r_{41} = & -x_{12}^2-8x_{12}x_{54}-4x_{12}x_{55}-16x_{54}^2-16x_{54}x_{55}-4x_{55}^2 \\
& +x_{56} \\
r_{42} = & -x_{10}-x_{10}^2-4x_{10}x_{12}-4x_{10}x_{56}-4x_{12}-4x_{12}^2 \\
& -8x_{12}x_{56}-4x_{56}-4x_{56}^2+x_{57} \\
r_{43} = & -10-12x_{57}-4x_{57}^2+x_{58} \\
r_{44} = & 1-4x_{12}-4x_{12}^2+8x_{12}x_{25}+16x_{12}x_{57}+8x_{12}x_{58}+4x_{25} \\
& -4x_{25}^2-16x_{25}x_{57}-8x_{25}x_{58}+8x_{57}-16x_{57}^2-16x_{57}x_{58} \\
& +4x_{58}-4x_{58}^2+x_{59} \\
r_{45} = & x_{11}-x_{11}^2+8x_{11}x_{12}-8x_{11}x_{25}-4x_{11}x_{59}-16x_{12}^2 \\
& +32x_{12}x_{25}+16x_{12}x_{59}-16x_{25}^2-16x_{25}x_{59}-4x_{59}^2+x_{60} \\
r_{46} = & -10-12x_{52}-4x_{52}^2+x_{61} \\
r_{47} = & 1-16x_{52}^2-16x_{52}x_{60}-16x_{52}x_{61}-4x_{60}^2-8x_{60}x_{61} \\
& -4x_{61}^2+x_{62} \\
r_{48} = & x_{12}-x_{12}^2-8x_{12}x_{60}-4x_{12}x_{62}-16x_{60}^2-16x_{60}x_{62} \\
& -4x_{62}^2+x_{63} \\
r_{49} = & 4x_{10}-4x_{10}^2-16x_{10}x_{11}+4x_{10}x_{12}+8x_{10}x_{21}+8x_{11} \\
& -16x_{11}^2+8x_{11}x_{12}+16x_{11}x_{21}-x_{12}-x_{12}^2-4x_{12}x_{21} \\
& -4x_{21}-4x_{21}^2+x_{64} \\
r_{50} = & 1-4x_{10}-4x_{10}^2-16x_{10}x_{11}+8x_{10}x_{21}+8x_{10}x_{64}-8x_{11} \\
& -16x_{11}^2+16x_{11}x_{21}+16x_{11}x_{64}+4x_{21}-4x_{21}^2-8x_{21}x_{64} \\
& +4x_{64}-4x_{64}^2+x_{65} \\
r_{51} = & -2+4x_{10}-16x_{10}^2-64x_{10}x_{11}+32x_{10}x_{21}+16x_{10}x_{63} \\
& +16x_{10}x_{65}+8x_{11}-64x_{11}^2+64x_{11}x_{21}+32x_{11}x_{63} \\
& +32x_{11}x_{65}-4x_{21}-16x_{21}^2-16x_{21}x_{63}-16x_{21}x_{65} \\
& -4x_{63}^2-8x_{63}x_{65}-2x_{65}-4x_{65}^2+x_{66} \\
r_{52} = & -2-4x_{66}-4x_{66}^2+x_{67} \\
r_{53} = & -x_3x_{15}+x_{68} \\
r_{54} = & x_{14}-x_{15}-2x_{24}x_{68}-x_{69} \\
r_{55} = & x_{14}-x_{15}-4x_{26}x_{68}-2x_{29}x_{68}-x_{70} \\
r_{56} = & x_{14}-x_{15}-2x_{31}x_{68}-6x_{68}-x_{71} \\
r_{57} = & x_{14}-x_{15}-4x_{32}x_{68}-2x_{34}x_{68}-x_{72} \\
r_{58} = & x_{14}-x_{15}-2x_{36}x_{68}+6x_{68}-x_{73} \\
r_{59} = & x_{10}x_{68}+x_{11}x_{68}+x_{14}-x_{15}+4x_{21}x_{68}+2x_{24}x_{68} \\
& +4x_{26}x_{68}+2x_{29}x_{68}+2x_{31}x_{68}+4x_{32}x_{68}+2x_{34}x_{68} \\
& +2x_{36}x_{68}+2x_{41}x_{68}+2x_{45}x_{68}+4x_{66}x_{68}+2x_{67}x_{68} \\
& -x_{68}x_{102}-x_{74} \\
r_{60} = & x_{14}-x_{15}-2x_{41}x_{68}-x_{75} \\
r_{61} = & x_{14}-x_{15}-2x_{45}x_{68}-4x_{68}-x_{76}
\end{aligned}
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\begin{aligned}
r_{62} =& -x_{10}x_{11} + x_{10}x_{12} + x_{11} + x_{11}x_{12} - x_{12} - x_{12}^2 + x_{77}\\
r_{63} =& \ 1 + 2x_{13} + x_{13}^2 - x_{77}^2 + x_{78}\\
r_{64} =& -x_{78}^2 + x_{79}\\
r_{65} =& -4x_{66}x_{68} - 2x_{67}x_{68} + 2x_{68} + x_{80}\\
r_{66} =& -x_{14}^2 + 2x_{14}x_{15} + 2x_{14}x_{80} - x_{15}^2 - 2x_{15}x_{80} - x_{80}^2\\
&+ x_{81}\\
r_{67} =& -1 - x_3x_{11} + x_{82}\\
r_{68} =& -x_{15}x_{82} + x_{84}x_{85} + x_{86}\\
r_{69} =& -1 + x_3 - x_3x_{10} + x_{83}\\
r_{70} =& -1 + 2x_3 + 2x_3x_{10} + 4x_3x_{11} - 2x_3x_{21} + x_{84}\\
r_{71} =& -x_{14}x_{83} + x_{85}\\
r_{72} =& -6x_3x_{15} + x_{14}x_{82} + 12x_{15} - 8x_{15}x_{82} - 4x_{15}x_{83} - x_{15}x_{84}\\
&+ x_{87}\\
r_{73} =& -x_{14}x_{69} + x_{15}x_{69} + x_{88}\\
r_{74} =& -x_{70}x_{88} + x_{89}\\
r_{75} =& -x_{71}x_{89} + x_{90}\\
r_{76} =& -x_{72}x_{90} + x_{91}\\
r_{77} =& -x_{73}x_{91} + x_{92}\\
r_{78} =& -x_{74}x_{92} + x_{93}\\
r_{79} =& -x_{75}x_{93} + x_{94}\\
r_{80} =& -x_{76}x_{94} + x_{95}\\
r_{81} =& -x_{86}x_{95} + x_{96}\\
r_{82} =& -x_{87}x_{96} + x_{97}\\
r_{83} =& -x_{97}^2 + x_{98}\\
r_{84} =& -x_2 - x_2x_{18} + x_{18} + x_{99}\\
r_{85} =& -x_{16}x_{99} + x_{100}\\
r_{86} =& -2x_1^2 + 2x_1x_{14} + 2x_1x_{15} + 2x_1x_{100} + 2x_1x_{101} - x_{14}^2\\
&- 2x_{14}x_{100} - x_{15}^2 - 2x_{15}x_{101} + x_{19}x_{103} - x_{79}x_{98} - x_{81}x_{98}\\
&- x_{100}^2 - x_{101}^2\\
r_{87} =& -x_2x_{17} + x_{17} - x_{17}x_{99} + x_{101}\\
r_{88} =& -x_{106} - 2x_{106}x_{237} + x_{108}\\
r_{89} =& -x_5x_{130} + x_{104}x_{228} - x_{107} - x_{108}x_{130}\\
r_{90} =& \ x_{109}x_{229} - x_{111}x_{133} - x_{112}\\
r_{91} =& \ x_{113}x_{229} - x_{114}x_{133} - x_{115}\\
r_{92} =& \ x_{116}x_{229} - x_{117}x_{133} - x_{118}\\
r_{93} =& \ x_{119}x_{229} - x_{120}x_{133} - x_{121}\\
r_{94} =& \ x_{122}x_{230} - x_{124}x_{133} - x_{125}\\
r_{95} =& \ x_{126}x_{230} - x_{127}x_{133} - x_{128}\\
r_{96} =& -x_{19}x_{130} + x_{133}\\
r_{97} =& \ x_{129} - x_{131}x_{133} - x_{132}\\
r_{98} =& -x_{133}x_{135} + x_{134} - x_{136}
\end{aligned}
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\begin{aligned}
r_{99} &= -x_{133}x_{138} + x_{137}x_{228} - x_{139}\\
r_{100} &= -x_{133}x_{141} + x_{140}x_{229} - x_{142}\\
r_{101} &= x_{143} - 2x_{143}x_{145} + x_{146}\\
r_{102} &= -x_{143}x_{146} - x_{146} + x_{147}\\
r_{103} &= 2x_{144} + x_{144}^2 + x_{146}^2 - x_{147}^2\\
r_{104} &= -x_{144}x_{149} - x_{149} + x_{150}\\
r_{105} &= 2x_{148} + x_{148}^2 + x_{149}^2 - x_{150}^2\\
r_{106} &= x_4^2 - 2x_4x_{144} - 2x_{144} + x_{151}\\
r_{107} &= -1 - x_4x_{149} + x_6 - x_{148} - x_{149} + x_{150} + x_{151}x_{152}\\
r_{108} &= 1 - 2x_5 + x_5^2 - 2x_5x_{144} + x_{153}\\
r_{109} &= -x_5x_{198} + x_{19} + x_{153}x_{154} - x_{197} + x_{199}\\
r_{110} &= -1 + 2x_{144}x_{213} - x_{155} + 2x_{213} - x_{213}^2\\
r_{111} &= 1 + x_{155}x_{156} - x_{197} - x_{198}x_{213} + x_{199} + 2x_{237}\\
r_{112} &= -x_{144}x_{158} - x_{158} + x_{159}\\
r_{113} &= 2x_{157} + x_{157}^2 + x_{158}^2 - x_{159}^2\\
r_{114} &= 1 - 4x_{144} - 4x_{144}x_{237} + x_{160} + 4x_{237} + 4x_{237}^2\\
r_{115} &= -1 + x_{104} - x_{157} - 2x_{158} - 2x_{158}x_{237} + x_{159} + x_{160}x_{161}\\
r_{116} &= -1 + x_{105} - 2x_{144}x_{162} - x_{148} - x_{149} - 2x_{149}x_{237} + x_{150} + x_{160}x_{162}\\
&\quad + x_{162} + 4x_{162}x_{237}\\
r_{117} &= 1 + x_{144} - x_{144}x_{164} - x_{164} + x_{165}\\
r_{118} &= 1 + 2x_{163} + x_{163}^2 - 2x_{164} + x_{164}^2 - x_{165}^2\\
r_{119} &= 1 + x_{109} + x_{160}x_{166} - x_{163} - 2x_{164} - 2x_{164}x_{237} + x_{165} + 2x_{237}\\
r_{120} &= -2x_{144} - x_{144}^2 + x_{167}\\
r_{121} &= -1 + x_{110} + x_{144}x_{150} - 2x_{144}x_{168} - x_{148} - 2x_{148}x_{237} - x_{149}x_{167}\\
&\quad - 2x_{150}x_{237} + x_{160}x_{168} + x_{168} + 4x_{168}x_{237} - 2x_{237}\\
r_{122} &= -x_{144}x_{170} - x_{170} + x_{171}\\
r_{123} &= 2x_{169} + x_{169}^2 + x_{170}^2 - x_{171}^2\\
r_{124} &= -1 + x_{113} + x_{160}x_{172} - x_{169} - 2x_{170} - 2x_{170}x_{237} + x_{171}\\
r_{125} &= -x_{144}x_{174} - x_{174} + x_{175}\\
r_{126} &= 2x_{173} + x_{173}^2 + x_{174}^2 - x_{175}^2\\
r_{127} &= -1 + x_{116} + x_{160}x_{176} - x_{173} - 2x_{174} - 2x_{174}x_{237} + x_{175}\\
r_{128} &= -x_{144}x_{178} - x_{178} + x_{179}\\
r_{129} &= 2x_{177} + x_{177}^2 + x_{178}^2 - x_{179}^2\\
r_{130} &= -1 + x_{119} + x_{160}x_{180} - x_{177} - 2x_{178} - 2x_{178}x_{237} + x_{179}\\
r_{131} &= -x_{144}x_{182} - x_{182} + x_{183}\\
r_{132} &= 2x_{181} + x_{181}^2 + x_{182}^2 - x_{183}^2\\
r_{133} &= -1 + x_{122} + x_{160}x_{184} - x_{181} - 2x_{182} - 2x_{182}x_{237} + x_{183}\\
r_{134} &= -x_{144}x_{186} - x_{186} + x_{187}\\
r_{135} &= 2x_{185} + x_{185}^2 + x_{186}^2 - x_{187}^2
\end{aligned}
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\begin{aligned}
r_{136} =& -1 + x_{123} - 2x_{144}x_{188} + x_{160}x_{188} - x_{185} - x_{186} - 2x_{186}x_{237} + x_{187} \\
&+ x_{188} + 4x_{188}x_{237} \\
r_{137} =& -x_{144}x_{190} - x_{190} + x_{191} \\
r_{138} =& 2x_{189} + x_{189}^2 + x_{190}^2 - x_{191}^2 \\
r_{139} =& -1 + x_{126} + x_{160}x_{192} - x_{189} - 2x_{190} - 2x_{190}x_{237} + x_{191} \\
r_{140} =& -x_{144}x_{194} - x_{194} + x_{195} \\
r_{141} =& 2x_{193} + x_{193}^2 + x_{194}^2 - x_{195}^2 \\
r_{142} =& -1 + x_{129} + x_{160}x_{196} - x_{193} - 2x_{194} - 2x_{194}x_{237} + x_{195} \\
r_{143} =& 1 - x_{144}x_{198} - x_{198} + x_{199} \\
r_{144} =& -1 + 2x_{197} + x_{197}^2 + x_{198}^2 - 2x_{199} - x_{199}^2 \\
r_{145} =& x_{130} - 2x_{144}x_{200} + x_{160}x_{200} - x_{197} - x_{198} - 2x_{198}x_{237} + x_{199} \\
&+ x_{200} + 4x_{200}x_{237} \\
r_{146} =& -x_{144}x_{202} - x_{202} + x_{203} \\
r_{147} =& 2x_{201} + x_{201}^2 + x_{202}^2 - x_{203}^2 \\
r_{148} =& -1 + x_{134} + x_{160}x_{204} - x_{201} - 2x_{202} - 2x_{202}x_{237} + x_{203} \\
r_{149} =& -x_{144}x_{206} - x_{206} + x_{207} \\
r_{150} =& 2x_{205} + x_{205}^2 + x_{206}^2 - x_{207}^2 \\
r_{151} =& -1 + x_{137} + x_{160}x_{208} - x_{205} - 2x_{206} - 2x_{206}x_{237} + x_{207} \\
r_{152} =& -x_{144}x_{210} - x_{210} + x_{211} \\
r_{153} =& 2x_{209} + x_{209}^2 + x_{210}^2 - x_{211}^2 \\
r_{154} =& -1 + x_{140} + x_{160}x_{212} - x_{209} - 2x_{210} - 2x_{210}x_{237} + x_{211} \\
r_{155} =& -10 - 2x_0 - 3x_0^2 - 2x_0x_4 - x_1^2 - x_2^2 - 2x_4 \\
&- 2x_4^2 - x_5^2 - x_6^2 + 2x_{10} - x_{10}^2 - x_{11}^2 - x_{12}^2 \\
&- x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 - x_{17}^2 - x_{18}^2 - x_{19}^2 \\
&- x_{102}^2 - x_{104}^2 - x_{105}^2 - x_{109}^2 - x_{110}^2 - x_{113}^2 - x_{116}^2 \\
&- x_{119}^2 - x_{122}^2 - x_{123}^2 - x_{126}^2 - x_{129}^2 - x_{130}^2 - x_{134}^2 \\
&- x_{137}^2 - x_{140}^2 + x_{143} - x_{213}^2 - 12x_{237} - 8x_{237}^2 \\
r_{156} =& -x_5x_{10} - x_5x_{11} - x_5x_{12} - x_5x_{13} - x_5x_{14} - x_5x_{15} \\
&- x_5x_{16} - x_5x_{17} - x_5x_{18} + x_{213} \\
r_{157} =& -16 + x_0 - x_{214}^2 \\
r_{158} =& -1 + x_2 + x_6 - x_6^2 \\
r_{159} =& -x_{144}x_{216} - x_{216} + x_{224} \\
r_{160} =& -1 + x_{215}^2 + x_{216}^2 - x_{224}^2 \\
r_{161} =& -x_{218} - x_{218}x_{219} + x_{225} \\
r_{162} =& -1 + x_{217}^2 + x_{218}^2 - x_{225}^2 \\
r_{163} =& -x_{210}^2 + x_{226} \\
r_{164} =& x_{216} - 2x_{220}x_{226} - x_{226} \\
r_{165} =& x_{144}x_{215} + x_{221} \\
r_{166} =& -x_{144} - x_{215}x_{221} + x_{219} \\
r_{167} =& -1 - x_{209} - x_{215}x_{222} + x_{217} \\
r_{168} =& -x_{102} - x_{210}x_{223} + x_{218}
\end{aligned}
$$

$$
\begin{aligned}
r_{169} = {} & 2x_0^2 + 2x_0x_4 - 2x_0x_{149} - 2x_0x_{158} + x_1^2 - 2x_1x_{198} \\
& + x_2^2 - 2x_2x_{186} + x_4^2 - 2x_4x_{158} + x_{10}^2 - 2x_{10}x_{164} \\
& + x_{11}^2 - 2x_{11}x_{170} + x_{12}^2 - 2x_{12}x_{174} + x_{13}^2 - 2x_{13}x_{178} \\
& + x_{14}^2 - 2x_{14}x_{182} + x_{15}^2 - 2x_{15}x_{190} + x_{16}^2 - 2x_{16}x_{194} \\
& + x_{17}^2 - 2x_{17}x_{202} + x_{18}^2 - 2x_{18}x_{206} + x_{102}^2 - 2x_{102}x_{210} \\
& - x_{144}x_{227} + x_{148}^2 + x_{149}^2 + x_{157}^2 + x_{158}^2 + x_{163}^2 + x_{164}^2 \\
& + x_{169}^2 + x_{170}^2 + x_{173}^2 + x_{174}^2 + x_{177}^2 + x_{178}^2 + x_{181}^2 \\
& + x_{182}^2 + x_{185}^2 + x_{186}^2 + x_{189}^2 + x_{190}^2 + x_{193}^2 + x_{194}^2 \\
& + x_{197}^2 + x_{198}^2 + x_{201}^2 + x_{202}^2 + x_{205}^2 + x_{206}^2 + x_{209}^2 \\
& + x_{210}^2 \\
r_{170} = {} & - x_{105}x_{228} + x_{130} \\
r_{171} = {} & - x_{110}x_{229} + x_{130} \\
r_{172} = {} & - x_{123}x_{230} + x_{130} \\
r_{173} = {} & - 2 - x_{102}^2 - x_{149}^2 - x_{158}^2 + 2x_{164} - x_{164}^2 - x_{170}^2 - x_{174}^2 \\
& - x_{178}^2 - x_{182}^2 - x_{186}^2 - x_{190}^2 - x_{194}^2 - x_{198}^2 - x_{202}^2 \\
& - x_{206}^2 + x_{210}^2 - x_{231}^2 - x_{232}^2 - x_{233} - x_{233}^2 \\
r_{174} = {} & - 1 - 4x_{107}^2 - 4x_{112}^2 - 4x_{115}^2 - 4x_{118}^2 - 4x_{121}^2 - 4x_{125}^2 \\
& - 4x_{128}^2 + x_{130}^2 - 4x_{132}^2 - 4x_{136}^2 - 4x_{139}^2 - 4x_{142}^2 - x_{234}^2 \\
& - x_{235}^2 - x_{236} - x_{236}^2 - 4x_{239}^2 \\
r_{175} = {} & - x_5x_{238} + x_{130} \\
r_{176} = {} & x_3x_{130} - x_6x_{238} - x_{239}
\end{aligned}
$$

B Machine-readable specification and exact evaluation

Each equation record declares its domain and ordered coordinates. For a sum-of-squares profile, evaluate every quadratic residual and add its square. For a Gram profile, also add each signed mixed product in `gram_cross_terms`. The collected coefficient file is a second exact representation of the same polynomial. The verifier expands the residual expression and compares it to that full coefficient list.

No externally supplied semantic witness is needed to evaluate a tuple. The evaluators are not solvers. A result at any particular tuple gives no decision procedure for the existence of an arbitrary integer zero.

References

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